

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCA
Course: Database Management System (Old)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define database and DBMS. What are the advantages of using DBMS over file management system? 7
b) Draw an ER diagram for the “**online food ordering system**” using extended features like generalization and specialization. 8
2. a) What are Set Operations in DBMS? Explain UNION and INTERSECTION. 7
b) Consider the following database schema: 8
Students (student_id, name, dob, gender)
Courses (course_id, course_name, credits)
Instructors (instructor_id, name, department)
Enrollments (enrollment_id, student_id, course_id, instructor_id, enrolled_date)
Write SQL statements for the following queries:
 - i. List all courses a student named ‘Sujan’ is enrolled in.
 - ii. Find out the number of students enrolled on each course.
 - iii. Find the average number of credits taken by students.
 - iv. Show the course names starting with ‘J’.
3. a) What is a View? Explain the applications of a view in DBMS. 7
b) Convert the following 2NF relation into 3NF (Consider name as PK) 8

<u>Name</u>	Address	Phone	Salary	Post
A	KTM	11111	20000	Engineer
B	BKT	22222	20000	Engineer
C	KTM	11111	20000	Engineer
D	BKT	22222	15000	Overseer
E	KTM	11111	10000	Officer

Also, check if any insertion, updation, or deletion anomaly occurs or not? if yes, then give an example.

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| 4. | a) | What are the constraints, and why do we need them? Explain all the types of integrity constraints with an example. | 7 |
| | b) | What is query optimization? Explain the steps involved in query processing. | 8 |
| 5. | a) | Differentiate between authentication and authorization. How encryption and decryption occur in private key and public key cryptography? | 7 |
| | b) | What are the causes of failure? Explain log-based recovery with an example. | 8 |
| 6. | a) | Define indexing. Explain index sequential file organization with advantage and disadvantages. | 7 |
| | b) | Define concurrency control. How do two phase locking protocol control concurrency? | 8 |
| 7. | | Write short notes on: (Any two) | 2×5 |
| | a) | Object Relational Model (ORM) | |
| | b) | Hash Index | |
| | c) | Data abstraction | |